

UNIVERSITY EXAMINATION

SUBJECT: ADVANCED ALGORITHMS | SEMESTER 7

Duration: 3 Hours | Max Marks: 80

Instructions: Attempt all questions. Assume suitable data if necessary.

- Q1(a) Describe architectural principles and implementation of Convolutional Neural Networks for image classification in com
- Q1(b) Describe architectural principles and implementation of Recurrent Neural Networks and Vanishing Gradient Problem
- Q1(c) Describe architectural principles and implementation of Long Short Term Memory Networks and Gated Recurrent Uni
- Q1(d) Describe architectural principles and implementation of Autoencoders Denoising and Variational Architectures in com
- Q1(e) Describe architectural principles and implementation of Generative Adversarial Networks Generator Discriminator Los
- Q2(a) Describe architectural principles and implementation of Gradient Descent Optimization Adam Stochastic RMSprop in
- Q2(b) Describe architectural principles and implementation of Backpropagation Error Signal Chain Rule Derivatives in comp
- Q2(c) Describe architectural principles and implementation of Activation Functions ReLU Sigmoid Softmax Tanh Properties
- Q2(d) Describe architectural principles and implementation of Regularization Techniques L1 L2 Dropout Weight Decay in co
- Q2(e) Describe architectural principles and implementation of Transfer Learning Fine Tuning Pretrained Backbone Models i